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Notes for Abstract Algebra

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Chapter 1

Structure of Groups

1.1 Sylow Theory

1.1.1 Group Actions

A (left) action of group G on a set Ω is a morphism $G \rightarrow \text{Sym}(\Omega)$, and a right action is $G^{op} \rightarrow \text{Sym}(\Omega)$. Since there is a natural morphism $G \rightarrow G^{op}$ by inverse, a left action induces a right action and vice versa.

Stablizer-Orbits: An action can be written as direct product of surjective actions, consider the orbits decomposition $\Omega = \coprod_x \Omega_x$ where $\Omega_x = Gx$ is G -invariant subset, then this action is also a map $G \rightarrow \prod_x \text{Sym}(\Omega_x) \hookrightarrow \text{Sym}(\Omega)$. If we consider the action $G \rightarrow \sum(\Omega_x)$, and let G_x be the stablizer of x , there is an isomorphism $\Omega_x \cong G/G_x$ of G -sets. There is also classification theorem of such indecomposable actions: $G/H \cong G/K$ iff H is conjugate to K . In fact, the stablizers in the same orbits is conjugate to each other by $(G_{gx})^g = G_x$.

Remark: In particular, if we consider a permutation $\sigma \in S_n$ and $\langle \sigma \rangle \hookrightarrow S_n$, this decomposition coincides with the decomposition of σ into cycles.

If we have a group G , we can construct several useful actions. G acts on all right cosets of a subgroup H by right multiplication (In particular, it acts on $\Omega = G$ as the cosets of 1.) The stablizer of Hx is H^x , and the kernel of this action is $\text{core}_G H = \bigcap H^x$. The core of H in G is exactly the unique largest normal subgroup of G contained in H , since it contains all others. We have an monomorphism $G/\text{core}_G H \rightarrow \text{Sym}(H \backslash G)$. As a corollary, if G has a subgroup of index n , then there is a normal subgroup, i.e. the core of this group, with index some factor of $n!$. If G is simple and contains a subgroup of index n , then G can be imbedded in S_n .

Another interesting action is the restriction of above action on H , the fixed points of this action are exactly cosets $Hx, x \in N_G(H)$. In particular, if H is a p -subgroup, then $[G : H] = |H \backslash G| = |H \backslash N_G(H)| = [N_G(H) : H] \pmod p$.

and G can act on all conjugates of a subset of Ω by conjugation.

1.1.2 Sylow Theorems

Theorem 1.1.1 (Cauchy)

Let G be a finite group and p be a prime. If p divides the order of G , then G has an element of order p .

Proof. Consider the set Ω of n -tuples $\{x_j\}$ s.t. $x_1x_2\cdots x_n = 1$, clearly $|\Omega| = p^{n-1}$. $\mathbb{Z}_p$ acts on this set, and each orbits must have length p or 1, but this action has fixed points $\{(x) : x^p = 1\}$, thus there are other $mp - 1$ elements of order p . We can also conclude that **the number of cyclic group of order p is 1 mod p** . $\square$

Remark: There is also another way to prove the theorem. First the abelian version of the theorem is easy: if $|G| = pn$, we use induction on n . There is an element a of order m , if $p \nmid m$ then $m \mid n$, $G/\langle a \rangle$ has an element $x\langle a \rangle$ of order p by induction, and since p must divides the order of x , we're done.

For nonabelian case, note that if p does not divide the length center Z of G , then we have some orbits x^G of length $[G : C_G(x)]$ that is not divisible by p , hence $C_G(x)$ must be a proper subgroup whose order is divisible by p . Now we can use induction on n if $|G| = pn$.

Theorem 1.1.2 (Sylow I)

Let G be a finite group, and let p be a prime. Then G has a subgroup of order p^n if $p^n \mid |G|$. In particular, G has a Sylow p -subgroup.

Proof. If have subgroup H of order p^{n-1} , $[G : H] = [N_G(H) : H]$ modulo p . Thus $N_G(H) \neq H$, and $N_G(H)/H$ must have $(1 \bmod p)$ many subgroups of order p , these pulls back subgroups of order p^n s.t. H is normal in it. Continuing this induction, we can find a subnormal series from 1 to a Sylow p -subgroup of G . $\square$

Remark: There is another method (Wielandt) of proving the theorem, if $|G| = p^am$, we consider the class Ω all subsets of cardinality p^a . Note that modulo p we have

$$(1 + X)^{p^am} = (1 + X^{p^a})^m = \cdots + mX^{p^a} + \cdots$$

hence

$$|\Omega| = \binom{p^am}{p^a} \equiv m \pmod{p}$$

When $p \nmid m$, we must have some orbits $\mathcal{O}$ with $p \nmid |\mathcal{O}|$. If G_S stabilizes $S \in \Omega$, by orbit-stabilizer theorem we have $[G : G_S] = |\mathcal{O}| \not\equiv 0 \pmod{p} \Rightarrow p^a \mid |G_S|$. But $|G_S| = |sG_S| \leq |S| = p^a$, so $|G_S| = p^a$. Note that first Sylow theorem 1.1.2 easily implies Cauchy's theorem 1.1.1.

We denote the set of Sylow p -subgroups by $\text{Syl}_p(G)$, and $n_p = |\text{Syl}_p(G)|$.

Theorem 1.1.3 (Sylow II)

If H is a p subgroup, and P is some Sylow p -subgroup, then there is $a \in G$ s.t. $H \subset P^a$.
As a consequence, Sylow subgroups are conjugate with each other.

Proof. Let P act on $H \backslash G$, then the fixed points are $\{xP : H \subset P^x\}$ and the number of them must not be divisible by p hence cannot be 0. $\square$

Theorem 1.1.4 (Sylow III)

$n_p = 1 \pmod p$ and divides $|G/P|$.

Proof. Sylow subgroups are one orbit of conjugate action of G on subgroups, so its length is $|G : N_G(P)|$. Let P acts on $\text{Syl}_p(G)$ by conjugation, the fixed points are $\{Q : P \subset N_G(Q)\}$. But $P, Q \in \text{Syl}_p(N_G(Q))$ so $P = Q$ and P is the only fixed point. $\square$

Remark: There is a stronger version in Martin Isaacs. Also, the number of p -subgroups of fixed order is also congruent to 1 mod p . See [this MSE post](#), [this MSE post](#) and [this post](#)

Corollary 1.1.5

For $P \in \text{Syl}_p(G)$, $N = N_G(P)$ is self-normalizing.

Proof. P is the only Sylow p -subgroup of N , if $N^x \subset N$ then $P^x \subset N \Rightarrow P^x = P \Rightarrow x \in N$. $\square$

Remark: There is yet another crazy method, using linear algebra, to prove the Sylow theorems. Let $|G| = p^a m = qm$. The idea is that by taking the adjoint representation of G over $\mathbb{F}_q$, we can imbed G into $\text{GL}_n(\mathbb{F}_q)$ which contains a Sylow p -subgroup as we shall see. (The representation theory does not work here since q divides the order of G .) Observe that

$$|\text{GL}_n(\mathbb{F}_q)| = q^{n(n-1)/2} \prod_{j=1}^n (q^j - 1)$$

, because we can select elements in the following way: first row can be anything but 0, so $q^n - 1$ many options, the $k + 1$ -th row can be anything but the linear combination of the former k rows, thus we must miss q^k vectors and left with $q^n - q^k$ options. In total we have

$$\prod_{j=0}^{n-1} (q^n - q^j) = q^{n(n-1)/2} \prod_{j=1}^n (q^j - 1)$$

possibilities. But immediately we see that the subgroup of upper triangular matrices also have $q^{n(n-1)/2}$ many elements, thus a p -subgroup of $\text{GL}_n(\mathbb{F}_q)$!

We now show that, if G has a Sylow p -subgroup P , then any subgroup H also has a Sylow p -subgroup. Let H acts on G/P , then there must be some orbits, say $H(xP)$, with length not divisible by p . But the stablizer Q of xP is $H \cap P^{x^{-1}}$ which is a p -subgroup. $|H|/|Q|$ not divisible by p implies $|Q|$ contains p parts of H , so Q must be a Sylow group.

See also A. J. Weir's original paper "[Sylow \$p\$ -subgroups of the general linear group over finite fields of characteristic \$p\$](#) ".

1.2 Semi-direct Product

1.3 Krull-Schmidt theorem

Consider the class of subgroups of G , ascending (descending) chain condition (ACC/DCC) is equivalent to maximal (minimal) condition. A group that satisfies ACC on subgroups is called a Noetherian group, and a group that satisfies DCC is called an Artinian group. ACCN/DCCN denotes ascending (descending) chain condition on normal subgroups.

Theorem 1.3.1

ACCN or DCCN implies G (and any nontrivial normal subgroup of G) is a finite direct product of indecomposable subgroups.

Proof. Consider the class

$$\mathcal{A} = \{1 \neq H \triangleleft G : H \text{ is not a finite direct product of indecomposable subgroups}\}$$

then $H \in \mathcal{A}$ implies H is decomposable, say $H = H_1 \times H_2$. At least one of the direct factors is again in $\mathcal{A}$, so $\mathcal{A}$ must be empty and $G \notin \mathcal{A}$. Now consider

$$\mathcal{B} = \{J : G = J \times H \text{ for some } H \in \mathcal{A}\}$$

then $J \in \mathcal{B}$ implies $J \times H_1 \in \mathcal{B}$, and DCCN implies there is a maximal subgroup N of $\mathcal{B}$, thus $\mathcal{B}$ is also empty and $1 \notin \mathcal{B}$, and $G \notin \mathcal{A}$. $\square$

We shall now investigate the uniqueness of this decomposition. The main result is

Theorem 1.3.2 (Krull-Schmidt)

Let G satisfy both ACCN and DCCN, and $G = G_1 \times \cdots \times G_s = H_1 \times \cdots \times H_t$ be two decomposition with each H_j, G_j indecomposable, then after reindexing $s = t, G_j \simeq H_j$ and

$$G = G_1 \times \cdots \times G_r \times H_{r+1} \times \cdots \times H_s$$

To prove the theorem we need some lemmas. Normal endomorphism are endomorphisms that commute with all inner automorphisms:

$$\text{Nor}(G) = \{f \in \text{End}(G) : f(a^b) = f(a)^b\}$$

Note that $f \in \text{Nor}(G)$ sends normal subgroup to normal subgroup.

Lemma 1.3.3

ACCN and $f \in \text{End}(G) \Rightarrow f \in \text{Aut}(G)$ iff f is epimorphism

DCCN and $f \in \text{Nor}(G) \Rightarrow f \in \text{Aut}(G)$ iff f is monomorphism.

Proof. Consider $1 \triangleleft \text{Ker} f \cdots \text{Ker} f^j \triangleleft \text{Ker} f^{j+1} \cdots$, ACCN implies there is n s.t. $\text{Ker} f^n = \text{Ker} f^{n+1}$. $f(a) = 1 \Rightarrow f^{n+1}(b) = 1$ for some $f^n(b) = a$, this implies $a = 1$ and $f \in \text{Aut}(G)$; Similarly, consider $G \cdots \text{Im} f^n \supset \text{Im} f^{n+1} \cdots$, DCCN implies there is m s.t. $\text{Im} f^m = \text{Im} f^{m+1}$, now $a \in G$, $f^m(a) = f^{m+1}(b) \Rightarrow f(b) = a$. $\square$

Lemma 1.3.4 (Fitting)

ACCN and DCCN implies $G = \text{Ker} f^n \times \text{Im} f^n$ for $f \in \text{Nor}(G)$, $n \gg 1$

Proof. For $n \gg 1$, $\text{Ker} f^n = \text{Ker} f^{n+j}$, $\text{Im} f^n = \text{Im} f^{n+j}$, $f^n(a) = f^{2n}(b)$, thus $a = af^n(b^{-1})f^n(b)$, where $f^n(af^n(b^{-1})) = 1$ and $f^n(b) \in \text{Im} f^n$. Besides, $a \in \text{Ker} f^n \cap \text{Im} f^n \Rightarrow a = f^n(b)$, $f^{2n}(b) = 1 \Rightarrow a = 1$. $\square$

$f \in \text{End}(G)$ is nilpotent if $f^n = 1$ for $n \gg 1$. $\text{Nil}(G)$ is in general not a submonoid. Lemma 1.6.8 shows that ,

Corollary 1.3.5

If G is indecomposable, then $\text{Nor}(G) \subset \text{Aut}(G) \sqcup \text{Nil}(G)$.

There is a point-wise product operation $f * g$ on $F(G, G)$, though this operation is not well-defined on $\text{End}(G)$. Note that we have distributive law $(f * g)h = fh * gh$, $h(f * g) = \varphi f * \varphi g$ if $\varphi \in \text{End}(G)$. The idea is, if f and g are nilpotent, then under some conditions, $f * g$ is also nilpotent.

Corollary 1.3.6

Let $G \neq 1$ be indecomposable and satisfies ACCN and DCCN. $f, g \in \text{Nor}(G) \cap \text{Nil}(G)$ and $f * g \in \text{End}(G)$, then $f * g \in \text{Nil}(G)$.

Proof. We prove by contradiction. Suppose $f * g \in \text{Aut}(G)$, let $\varphi = (f * g)^{-1}$, and $F = f\varphi$, $G = g\varphi$, then we shall show that $FG = GF$ and $F * G = G * F = \text{id}$ and thus by commutativity

$$\text{id} = (F * G)^n = *_k \binom{n}{k} F^k G^{n-k} = 1$$

for $n \gg 1$ since F, G is also nilpotent, contradicts the fact that $G \neq 1$. Indeed $(F * G)(x) = x \Rightarrow (G * F)(x^{-1}) = x^{-1} \Rightarrow F * G = G * F = \text{id}$, $F(F * G) = (F * G)F \Rightarrow FG = GF$. $\square$

We are now ready to prove Theorem 1.3.2.

Proof. We prove by induction. Suppose we have known there is reindex s.t. $G_j \simeq H_j$ for $j \leq r-1$ and $G = G_1 \times \cdots \times G_s = G_1 \times \cdots \times G_{r-1} \times H_r \times \cdots \times H_t$, let p_j be the projection map $G \rightarrow G_j \hookrightarrow G$ and q_j be the projection $G \rightarrow G_j \& H_j \hookrightarrow G$, then p_j, q_j is normal and $p_1 * p_2 \cdots p_s = q_1 * q_2 \cdots q_t = \text{id}$. Consider $p_r = p_r(q_1 * q_2 \cdots q_t) = p_r(q_r * \cdots q_t)$, and if we restrict each map on G_r (which also satisfies ACCN and DCCN), then at least one of $p_r q_j$ is an automorphism on G_r ! Thus we reindex

and let $p_r q_r \in \text{Aut}(G_r)$. $p_r(q_r p_r)^n q_r$ is an automorphism thus $q_r p_r$ is not nilpotent hence also an automorphism on H_r , and p_r, q_r gives $H_r \simeq G_r$.

To show the product $G_1 \cdots G_r H_{r+1} \cdots H_t$ is direct, note that $p_r q_r$ is automorphism on G_r and 1 on $G_1 \cdots G_{r-1} H_{r+1} \cdots H_t$ so $G_r \cap G_1 \cdots G_{r-1} H_{r+1} \cdots H_t = 1$.

Also the map $p_1 * \cdots p_{r-1} * p_r * q_{r+1} * \cdots q_t$ is a normal monomorphism, thus an automorphism. Thus $G = G_1 \times \cdots G_r \times H_{r+1} \times \cdots H_t$ □

1.4 Jordan-Hölder theorem

In abstract algebra, a composition series provides a way to break up an algebraic structure, such as a group or a module, into simple pieces. The need for considering composition series in the context of modules arises from the fact that many naturally occurring modules are not semi-simple, hence cannot be decomposed into a direct sum of simple modules. A composition series of a module M is a finite increasing filtration of M by submodules such that the successive quotients are simple and serves as a replacement of the direct sum decomposition of M into its simple constituents.

A composition series may not exist, and when it does, it need not be unique. Nevertheless, a group of results known under the general name Jordan-Hölder theorem asserts that whenever composition series exist, the isomorphism classes of simple pieces (although, perhaps, not their location in the composition series in question) and their multiplicities are uniquely determined. Composition series may thus be used to define invariants of finite groups and Artinian modules.

A **subnormal series** of a group G is a sequence of subgroups, each a normal subgroup of the next one. In a standard notation

$$1 = A_0 \triangleleft A_1 \triangleleft \cdots \triangleleft A_n = G.$$

If in addition each A_i is normal in G , then the series is called a normal series. The length of a series is the number of strict inclusions $A_i < A_{i+1}$, equivalently, the number of non-trivial factor groups. If the series has no repetition then the length is n .

A composition series is a maximal subnormal series (has no proper refinement). Equivalently, a subnormal series for which each of the A_i is a maximal normal subgroup of A_{i+1} . Equivalently, a composition series is a subnormal series for which each of the factor groups are simple.

Two composition series of a given group are equivalent if they have the same composition length and the same composition factors, up to permutation and isomorphism.

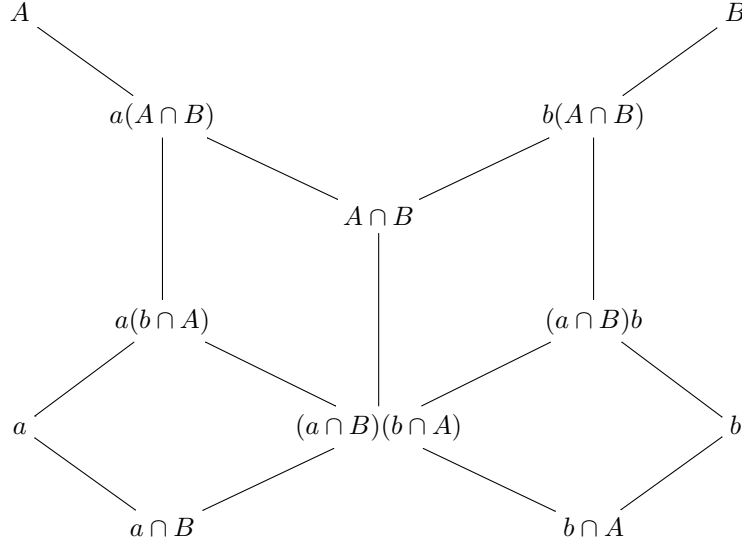
We shall investigate the uniqueness of the composition series under this equivalent relation. First we need a technical lemma (the butterfly lemma or Zassenhaus lemma) on the lattice of subgroups of a group or the lattice of submodules of a module:

Lemma 1.4.1 (Zassenhaus)

Suppose G is a group with subgroups A and B . Suppose $a \triangleleft A$ and $b \triangleleft B$ are normal

subgroups. Then there is an isomorphism of quotient groups:

$$\frac{a(A \cap B)}{a(A \cap b)} \cong \frac{(A \cap B)b}{(a \cap B)b}.$$



The lemma can be proved using the following generalization of second isomorphism theorem:

Proposition 1.4.2 (Benjamin Baumslag)

Let N be normal in K and K, L be subgroups of G . If $L^k = L$ for every $k \in K$ (i.e. K is in the normalizer of L). Then

$$\frac{KL}{NL} \cong \frac{K}{N(K \cap L)}$$

Proof. Recall that composition AB of groups A, B is group iff $AB = BA$. In our case $L^k = L$ thus both KL, NL are subgroups, and both K and L are in the normalizer of NL , so $NL \triangleleft KL$. Note that $N(K \cap L) \subset K \cap NL$, conversely, if $nl \in K$ for $n \in N \subset K, l \in L$, then $l \in K \Rightarrow nl \in N(K \cap L)$. By the second isomorphism theorem (here we consider everything as subgroups of KL):

$$\frac{K}{N(K \cap L)} = \frac{K}{(K \cap NL)} \cong \frac{K(NL)}{NL} = \frac{KL}{NL}$$

□

Now we can easily prove the lemma:

Proof. First note that $(A \cap b) \triangleleft (A \cap B)$, by the preceding proposition, $(A \cap b)a \triangleleft (A \cap B)a$

$$\frac{(A \cap B)a}{(A \cap b)a} \cong \frac{A \cap B}{(A \cap b)(a \cap B)}$$

A symmetric argument shows $(a \cap B)b \triangleleft a(A \cap B)$ and

$$\frac{(A \cap B)b}{(a \cap B)b} = \frac{A \cap B}{(A \cap b)(B \cap a)}$$

□

By intersecting the terms in one subnormal series with those in the other series, we prove the following Schreier refinement theorem:

Theorem 1.4.3 (Schreier)

Any two subnormal (or normal) series of subgroups of a given group have equivalent refinements.

Proof. Suppose that G_i, H_j are subnormal series, inserting $G_i(G_{i+1} \cap H_j)$ in $G_i \triangleleft G_{i+1}$, and similarly, $H_j(H_{j+1} \cap G_i)$ in $H_j \triangleleft H_{j+1}$. Note that

$$\frac{H_j(H_{j+1} \cap G_{i+1})}{H_j(H_{j+1} \cap G_i)} = \frac{G_i(G_{i+1} \cap H_{j+1})}{G_i(G_{i+1} \cap H_j)}$$

with denominator normal in numerator. The normal series case is trivial. □

As a consequence, if a composition series exists for a group G , then any subnormal series of G can be refined to a composition series that is equal to that composition series by inserting subgroups into the series up to maximality. In particular:

Theorem 1.4.4 (Jordan-Hölder)

Any two composition series of a given group are equivalent.

Every finite group has a composition series, but not every infinite group has one. For example, $\mathbb{Z}$ has no composition series.

Theorem 1.4.5

A group has finite composition length iff it satisfies ACC and DCC on subnormal subgroups

Proof. If G has composition length n , then every subnormal series must have length at most n . Conversely, we first choose a normal subgroup of G by ACC, then inductively choose maximal subnormal subgroups, by DCC we must stop at some simple group. □

In conclusion: a group has finite composition length if it satisfies the following equivalent conditions:

1. It possesses a composition series of finite length, viz., a subnormal series such that all the successive quotients are simple groups.
2. Every subnormal series (without repeated terms) can be refined to a composition series of finite length.
3. It satisfies both the ascending chain condition on subnormal subgroups and the descending chain condition on subnormal subgroups.

4. There is an upper bound on the length of any subnormal series for the group (this upper bound equals the composition length, i.e., the length of any composition series).

As for the case of modules: It is well known that a module has a finite composition series if and only if it is both an Artinian module and a Noetherian module. If R is an Artinian ring, then every finitely generated R -module is Artinian and Noetherian, and thus has a finite composition series. In particular, for any field k , any finite-dimensional module for a finite-dimensional algebra over k has a composition series, unique up to equivalence.

1.5 Solvable Group

A solvable group is a group that can be constructed from abelian groups using extensions.

A group G is called **solvable group** if it has a subnormal series $\{G_j\}$ whose factor groups G_{j+1}/G_j are all abelian (i.e. $[G_{j+1}, G_{j+1}] \leq G_j$), we call this series a solvable series. Or equivalently, if it has a **derived series** $\{D_j\}$ where $D_j = [D_{j+1}, D_{j+1}]$. Indeed,

$$D_{j+1} \leq G_{j+1} \Rightarrow D_j = [D_{j+1}, D_{j+1}] \leq [G_{j+1}, G_{j+1}] \leq G_j$$

thus the derived series is fastest descending series with Abelian quotients. The length of $\{D_j\}$ is called the derived length of the solvable group G .

Remark: Although $\{G_j\}$ not necessarily normal (D_8 for example), it give rise to a normal series D_j , so in the definition we can change subnormal to normal.

Theorem 1.5.1

The class of solvable groups is closed with respect to the formation of subgroups, images, and extensions of its members.

Proof. If we have a morphism f , a subgroup K and a solvable series G_j , $\{K \cap G_j\}$, $f(G_j)$ is subnormal with

$$[f(G_{j+1}), f(G_{j+1})] \leq f(G_j)$$

If we have a solvable series $\{N_i\}$ and $\{G_j/N\}$, then $\{N_i, G_j\}$. □

As a consequence

Corollary 1.5.2

The product NK of two solvable subgroups N, K with N normal of a group is solvable.

Proof. By second isomorphism theorem

$$\frac{NK}{N} = \frac{K}{N \cap K}$$

□

For a finite group, we have an equivalent definition using composition series.

Lemma 1.5.3

A refinement of solvable series is solvable.

Thus G is solvable, then the solvable series can be refined to a solvable composition series. But abelian simple group is cyclic of prime order, thus another definition for finite solvable group is:

Theorem 1.5.4

A finite group is solvable iff it has a composition series whose factors are cyclic of prime order.

Proof. We can also construct the series directly: we can fill in $G_j \leq G_{j+1}$ by a maximal normal proper subgroup $G_j \leq H_0 \leq G_{j+1}$, then do the same for $G_j \leq H_0$. In the end, H_{j+1}/H_j must be simple. $\square$

1.5.5 Example

1. Clearly every abelian group is solvable. If a group contains some simple nonabelian quotient or subgroup, then it's not solvable (Since solvable + simple implies abelian (of prime order)).
2. Dihedral group D_n is solvable, $p^a q^b$ -group is solvable (Burnside), S_3 is solvable but not nilpotent.
3. **The Feit-Thompson theorem** states that every finite group of odd order is solvable. In particular this implies that if a finite group is simple, it is either a prime cyclic or of even order.
4. Group G of order less than 120 is solvable except A_5 : All composition factors of G must be group smaller than A_5 , thus cyclic of prime order.
5. A group is said to be **perfect** if it equals its own commutator subgroup, or equivalently, if the group has no non-trivial abelian quotients (equivalently, its abelianization, which is the universal abelian quotient, is trivial). Such group is not solvable.

1.6 Nilpotent Group

Intuitively, a nilpotent group is a group that is "almost abelian". This idea is motivated by the fact that nilpotent groups are solvable, and for finite nilpotent groups, two elements having relatively prime orders must commute.

1.6.1 Central Series

If H, K are subgroups, commutator $[H, K] := \langle [h, k] : h \in H, k \in K \rangle$. Clearly $[H, K] = [K, H]$. Note that for normal subgroup K in another subgroup H , $[G, H] \leq K \iff H/K \leq Z(G/K)$, these conditions control the 'gaps' between K, H . Though the equality does not hold, if we fix H , we can ask the smallest such K and vice versa.

A **central series** is a subnormal series

$$1 = A_0 \triangleleft A_1 \triangleleft \cdots \triangleleft A_n = G$$

such that the successive quotients are central; that is, $[G, A_{i+1}] \leq A_i$, since $[G, A_{i+1}] \leq A_{i+1}$ implies A_{i+1} normal, we can rephrase the 'central' condition above as: each A_i is normal and $A_{i+1}/A_i \subseteq Z(G/A_i)$.

If a group has a central series, then there are two central series whose terms are extremal in certain senses. If we start from U_j , and let U_{j+1} be a bigger group s.t. $U_{j+1}/U_j = Z(G/U_j)$, Continuing in this way to choose the largest possible U_{j+1} given U_j produces what is called the **upper central series(UCS)**, and U_j is called the j -th center of G . We can also interpret this series reversely, i.e. we take the group G and "kill" it by repeatedly "killing" the center:

$$G \rightarrow G/Z(G) = G_1 \rightarrow G_1/Z(G_1) = G_2 \rightarrow G_2/Z(G_2) \cdots \rightarrow G_j/Z(G_j) = G_{j+1}$$

the pull back of $Z(G_{j+1})$ under the projection is exactly the pull back of $Z(G/U_j)$ under $G \rightarrow G/U_j$ if we assume the pull pack into G of $Z(G_j)$ is U_j .

Now if we start from L_{j+1} , and let L_j be the small group $L_j = [G, L_{j+1}]$, Continuing to choose L_i minimally given L_{i+1} produces what is called **the lower central series(LCS)**.

Clearly if there exists some central series A_j , then LCS L_j exists: if $L_{j+1} \leq A_{j+1}$, then $L_j = [G, L_{j+1}] \leq [G, A_{j+1}] = A_j$. Besides, if we know $A_j \leq U_j$, then $[A_{j+1}, G] \leq A_j \leq U_j \Rightarrow A_{j+1}/U_j \leq Z(G/U_j) \Rightarrow A_{j+1} \leq U_{j+1}$, thus UCS also exists.

Thus the existence of these central series are equivalent, and such group is called **nilpotent group**. For a nilpotent group, the lengths of the LCS and the UCS agree, which is the smallest n such that G has a central series of length n and is called the **nilpotency class** of G ; and G is said to be nilpotent of class n . (By definition, the length is n if there are $n+1$ different subgroups in the series, including the trivial subgroup and the whole group.)

If a group has nilpotency class at most n , then it is sometimes called a $\text{nil-}n$ group. It follows immediately from any of the above forms of the definition of nilpotency, that the trivial group is the unique group of nilpotency class 0, and groups of nilpotency class 1 are exactly the non-trivial abelian groups.

1.6.2 Some Elementary Facts

Proposition 1.6.1

If G is nilpotent, then every nontrivial normal subgroup N intersects $Z(G)$ nontrivially. As a corollary, a minimal normal subgroup of G is contained in the center.

Proof. If $U_{j+1} \cap N$ is the first nontrivial intersection, $[U_{j+1} \cap N, G] \leq [U_{j+1}, G] \cap N = 1$ hence $U_{j+1} \cap N \leq Z(G)$. $\square$

The proposition says if we let G act on N by conjugation, we must have a fixed point other than 1. If G is a p -group, this is easily seen by a simple calculation.

Lemma 1.6.2

For a nilpotent group G , a maximal normal abelian subgroup A (which exists by Zorn) is self-centralizing, i.e. $A = C_G(A)$.

Proof. The group $C_G(A)$ is normal in G since A is normal, thus there is nontrivial $gA \in C_G(A)/A \cap Z(G/A)$, now the abelian group $\langle A, g \rangle$ is normal since $[g, G] \subset A$. Contradicts the maximality of A . $\square$

Remark: This indicates the degree to which such subgroups control a nilpotent group: consider the conjugate action: $G \rightarrow \text{Aut} A$, with the kernel $C_G(A)$, if A is finite, then $|G/A| = |G/\text{Aut} A| \leq |\text{Aut} A|$, thus G must be finite.

If $A, C \leq G$ and $B, D \leq H$, we have a formula in $G \times H$:

$$[A \times B, C \times D] = [A, C] \times [B, D]$$

Theorem 1.6.3

The class of nilpotent groups is closed under the formation of subgroups, images, and finite direct products.

Proof. Let f be a morphism and K be a subgroup. If $\{A_j\}$ is a central series, $\{A_j \cap K\}$ is sub-normal series with $[(A_{j+1} \cap K), K] \leq (A_j \cap K)$ and $f(A_j)$ is normal in $f(G)$ with $[f(A_{j+1}), f(G)] \leq f(A_j)$.

If we have two nilpotent groups A, B with central series $\{A_i\}, \{B_j\}$, then $\{A_i, A \times B_j\}$ is a central series of $A \times B$ since

$$[A_j, A \times B] = [A_j, A] \leq A_{j+1}, [A \times B_j, A \times B] = [A, A] \times [B_j, B] \leq A \times B_{j-1}$$

$\square$

Remark: An extension of one nilpotent group by another need not be nilpotent -as S_3 shows: $[S_3, S_3] = A_3$ is not nilpotent but cyclic groups $\mathbb{Z}_2, \mathbb{Z}_3$ are. This is because if G/N and N is nilpotent and $\{N_i\}, \{G_j/N\}$ central series, $\{N_i\}$ is in general not central in G , although G_j is. There is, however, P. Hall's Criterion for such an extension to be nilpotent.

But we do have some specific case when pull back of a nilpotent group is also nilpotent. If N is central, then G/N nilpotent implies G is. One can see this by the 'killing' centers definition.

1.6.3 Characterizations of Finite Nilpotent Groups

There are several group-theoretical properties which for finite groups are equivalent to nilpotence. One of these is the normalizer condition, every proper subgroup is properly contained in its normalizer, i.e. it has no proper self-normalizing subgroup. Another such is the property that every maximal subgroup is normal. Here by a maximal subgroup we mean a proper subgroup which is

not contained in any larger proper subgroup. A subgroup K is subnormal if it appears as a term in some subnormal series of G (i.e. there is a subnormal series between K, G).

Recall that if P is Sylow p -subgroup of G , $K = N_G(K)$ for $N_G(P) \leq K$.

Theorem 1.6.4

Let G be a finite group. Then the following properties are equivalent:

- (i) G is nilpotent;
- (ii) every subgroup of G is subnormal;
- (iii) G satisfies the normalizer condition;
- (iv) every maximal proper subgroup of G is normal;
- (v) every Sylow subgroup is normal;
- (vi) G is the direct product of its Sylow subgroups.
- (vii) G has a normal subgroup of any order dividing $|G|$.

Proof. (i) $\Rightarrow$ (ii): If K is a subgroup, A_j is a central series of G , then KA_j is subnormal series between K, G .

(ii) $\Rightarrow$ (iii), (iii) $\Rightarrow$ (iv): trivial

(iv) $\Rightarrow$ (v) $\Rightarrow$ (vi), if P is Sylow p -subgroup of G , $N_G(P)$ is contained in some maximal subgroup K since G is finite, and this group must be normal and $K = N_G(K) = G$. Thus P must be normal. If $\{P_j\}$ are Sylow subgroups, then $P_i \cap P_j = 1$ and thus P_i commute with P_j . The element in $P_1 \cdots P_{i-1}, P_{i+1} \cdots P_s$ has order dividing $\prod_{j \neq i} |P_j|$. Thus P_j is direct product.

(vi) $\Rightarrow$ (vii) is trivial

(vii) $\Rightarrow$ (v) $\Rightarrow$ (vi) $\Rightarrow$ (i) since p -group is clearly nilpotent. $\square$

Remark: For finite group, we also have nilpotent iff every nontrivial homomorphic image of G has a nontrivial center. **add examples** and **to add torsion group** and semidirect product

Semidirect prod of nilpotent? Schur-Zassenhaus theorem

universal property?

Can any solvable finite group be obtained from abelian groups and combinations of taking subgroups, quotients, and semidirect products?

1.6.4 Tensor Products and P.Hall's Criterion

If we have the LCS $G_j, G_{j+1} = [G, G_j]$ and $G_1 = G$, then we have relations:

Lemma 1.6.5

$$[G_i, G_j] \leq G_{i+j}$$

Proof. If $i = 1$ this is trivial. The Hall-Witt identity

$$[[x, y^{-1}], z]^y \cdot [[y, z^{-1}], x]^z \cdot [[z, x^{-1}], y]^x = 1$$

shows that for normal subgroups A, B, C , then

$$[[A, B], C] \leqslant [[B, C], A][[C, A], B]$$

If $[G_{i-1}, G_j] \leqslant G_{i+j-1}$, then

$$[G_i, G_j] = [[G, G_{i-1}], G_j] \leqslant [G_{i+j}, G][G_{j+1}, G_{i-1}] \leqslant G_{i+j}$$

□

There is a well-defined pairing

$$G_j/G_{j+1} \times G/G_1 \rightarrow G_{j+1}/G_{j+2}; (aG_{j+1}, gG_1) \mapsto [a, g]G_{j+2}$$

since $[G_{j+1}, G] = G_{j+2}$, $[G_j, G_1] \leqslant [[G_j, G], G] = G_{j+2}$, $[G_j, G_1] \leqslant G_{j+2}$ thus well-defined. Besides, the pairing is bilinear

$$[ab, g] = [a, g] \cdot [[a, g], b] \cdot [b, g]$$

$$[a, gh] = [a, h] \cdot [a, g] \cdot [[a, g], h]$$

Let F_j be the j -th central factor G_j/G_{j+1} , thus F_1 be the abelianization. The pairing gives an epimorphism

$$F_j \otimes F_1 \rightarrow F_{j+1}$$

and inductively we have $F_1^{\otimes n} \rightarrow F_n$

Theorem 1.6.6

If P is a property inherited by images of tensor products and by extensions, then for nilpotent group G , if F_1 has property P , so is G .

For example, let P be the property of being finite. Then we obtain the result: if G is a nilpotent group and F_1 is finite, then G too is finite.

Theorem 1.6.7 (P.Hall)

If both a normal subgroup N and $G/[N, N]$ is nilpotent, so is G .

1.6.5 The Fitting Subgroup

Theorem 1.6.8 (Fitting)

If M and N are nilpotent normal subgroups of a group G , then their product MN is also a nilpotent normal subgroup of G ; if, moreover, M is nilpotent of class m and N is nilpotent of class n , then MN is nilpotent of class at most $m + n$.

In terms of order theory, (part of) Fitting's theorem can be stated as: The set of nilpotent normal subgroups form a lattice of subgroups. Thus the nilpotent normal subgroups of a finite group also form a bounded lattice, and have a top element, the Fitting subgroup.

However, nilpotent normal subgroups do not in general form a complete lattice, as a subgroup generated by an infinite collection of nilpotent normal subgroups need not be nilpotent, though it will be normal. The join of all nilpotent normal subgroups is still defined as the Fitting subgroup, but it need not be nilpotent.

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